CO-1-AT1

OSI Layer Calculations and Network Communication Analysis

Assignment Submission-1

Course: Computer Networks

Topic: Application of osi layers in enabling reliable data transmission across networks

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1. File Transfer using Transport and Data Link Layers

Introduction

The Transport Layer divides large files into smaller segments for reliable communication. The Data Link Layer encapsulates each segment into frames by adding headers and trailers for error detection and node-to-node delivery.

Given

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link Layer overhead = **40 bytes per frame**

Calculation

Step 1: Convert file size into bytes

150 MB = 150 × 1024 × 1024

= **157,286,400 bytes**

Step 2: Calculate number of segments

Number of Segments

= 157,286,400 ÷ 1500

= **104,858 segments**

Since one segment becomes one frame,

Number of Frames = 104,858

Step 3: Calculate Data Link Layer overhead

Overhead

= 104,858 × 40

= **4,194,320 bytes (≈ 4 MB)**

Step 4: Total transmitted data

157,286,400 + 4,194,320

= **161,480,720 bytes**

Convert into bits

161,480,720 × 8

= **1,291,845,760 bits**

Step 5: Transmission time

Transmission Time

= 1,291,845,760 ÷ 50,000,000

= **25.84 seconds**

Final Answer

- **Number of Segments = 104,858**
- **Number of Frames = 104,858**
- **Data Link Layer Overhead = 4,194,320 bytes**
- **Transmission Time = 25.84 seconds**

Explanation

The **Transport Layer** performs segmentation, sequencing, flow control, and error recovery to ensure reliable communication. The **Data Link Layer** adds headers and trailers, performs framing, detects errors using CRC, and provides reliable node-to-node transmission.

2. Sliding Window Protocol

Introduction

The Sliding Window Protocol allows multiple frames to be transmitted before receiving acknowledgements. It improves network utilization and reduces waiting time.

Given

- Window size = **6 frames**
- Total frames = **48**
- User data per frame = **1024 bytes**

Calculation

Number of windows

48 ÷ 6

= **8 windows**

Retransmissions

One frame is lost in every window.

Total Retransmissions

= **8 frames**

Final Answer

- **Transmission Windows = 8**
- **Retransmissions = 8**

Explanation

Flow control prevents the sender from overwhelming the receiver. It improves communication efficiency by avoiding congestion, reducing packet loss, and maintaining smooth data transmission.

3. IP Fragmentation

Introduction

When an IP packet is larger than the Maximum Transmission Unit (MTU), the Network Layer divides it into smaller fragments for successful transmission.

Given

- IP packet size = **12,000 bytes**
- MTU = **1500 bytes**
- IP Header = **20 bytes**

Calculation

Payload in each fragment

1500 − 20

= **1480 bytes**

Number of fragments

12,000 ÷ 1480

= 8.11

Rounded up,

= **9 fragments**

Header overhead

9×20

= **180 bytes**

Final Answer

- **Fragments Created = 9**
- **Header Overhead = 180 bytes**

Explanation

The **Network Layer** performs fragmentation, routing, logical addressing, and packet reassembly at the destination to ensure successful data delivery.

4. Sliding Window Protocol

Introduction

Sliding Window Flow Control improves transmission efficiency by allowing multiple frames to be transmitted before acknowledgements are received.

Calculation

Transmission Windows

$48 \div 6$

= **8**

Retransmissions

= **8**

Final Answer

- **Windows = 8**
- **Retransmissions = 8**

Explanation

Flow control prevents receiver buffer overflow and increases network throughput while ensuring reliable communication.

5. Session Layer Synchronization

Introduction

The Session Layer establishes, maintains, and synchronizes communication sessions. Synchronization points help resume communication after failures.

Given

- Video size = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Calculation

Number of checkpoints

$390 \div 60$

$= 6.5$

Completed checkpoints

= **6**

Checkpoint positions:

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

Data to retransmit

$390 - 360$

= **30 MB**

Final Answer

- **Checkpoints Created = 6**
- **Retransmitted Data = 30 MB**

Explanation

Synchronization checkpoints reduce retransmission time by allowing communication to resume from the latest checkpoint instead of restarting the complete transfer.

6. Presentation Layer Compression and Encryption

Introduction

The Presentation Layer performs data compression, encryption, and format conversion to improve transmission efficiency and data security.

Given

- Original file = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%**

Calculation

Compressed size

$900 \times 60\%$

= **540 MB**

Encrypted size

$540 \times 105\%$

= **567 MB**

Overall reduction

$900 - 567$

= **333 MB**

Percentage reduction

= (333 ÷ 900) × 100

= **37%**

Final Answer

- **Compressed Size = 540 MB**
- **Encrypted Size = 567 MB**
- **Overall Reduction = 37%**

Explanation

Compression reduces the amount of transmitted data, while encryption protects confidential information from unauthorized access.

7. FTP File Transfer

Introduction

FTP (File Transfer Protocol) is an Application Layer protocol used for reliable file transfer between computers.

Given

- Data = **3 GB**
- Throughput = **120 Mbps**
- Request size = **3 MB**

Calculation

Total requests

3 GB = 3072 MB

3072 ÷ 3

= **1024 requests**

Transmission time

3072 MB × 8

= 24,576 Mb

24,576 ÷ 120

= **204.8 seconds**

Final Answer

- **Requests Generated = 1024**
- **Transmission Time = 204.8 seconds**

Explanation

The Application Layer provides services such as FTP, email, web browsing, authentication, and user communication.

8. End-to-End Delay

Introduction

End-to-End Delay is the total time taken for a packet to travel from the sender to the receiver.

Given

Delay	Value
Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Calculation

Total Delay

4 + 3 + 5 + 2 + 1 + 18 + 12

= **45 ms**

Final Answer

Total End-to-End Delay = 45 ms

Explanation

- **Application Layer:** Provides network services to users.
- **Transport Layer:** Ensures reliable communication.
- **Network Layer:** Selects the best routing path.
- **Data Link Layer:** Performs framing and error detection.
- **Physical Layer:** Transmits bits over the medium.
- **Propagation Delay:** Time taken by signals to travel.
- **Transmission Delay:** Time required to send all bits.

9. Protocol Overhead and Efficiency

Introduction

Protocol overhead consists of additional bytes added by different OSI layers. Excessive overhead reduces network efficiency.

Given

- Useful Data = **80 MB**
- Frame Size = **1600 bytes**
- Overhead = **80 bytes**

Calculation

Payload per frame

1600 − 80

= **1520 bytes**

Useful data

80 MB

= **83,886,080 bytes**

Frames

83,886,080 ÷ 1520

= **55,189 frames**

Overhead

55,189 × 80

= **4,415,120 bytes**

Transmission efficiency

= (83,886,080 ÷ 88,301,200) × 100

= **95%**

Final Answer

- **Frames = 55,189**
- **Protocol Overhead = 4,415,120 bytes**
- **Transmission Efficiency = 95%**

Explanation

Although protocol overhead consumes bandwidth, it provides addressing, synchronization, sequencing, and error detection, which are essential for reliable communication.

10. Healthcare Network using Multiple OSI Layers

Introduction

In healthcare networks, different OSI layers work together to provide efficient, secure, and reliable transmission of sensitive medical records.

Given

- Original File = **240 MB**
- Compression = **30%**
- Segment Size = **1400 bytes**
- Frame Overhead = **60 bytes**

Calculation

Compressed file

240 × 70%

= **168 MB**

Convert to bytes

168 × 1024 × 1024

= **176,160,768 bytes**

Number of segments

176,160,768 ÷ 1400

= **125,830 segments**

Protocol overhead

125,830 × 60

= **7,549,800 bytes**

Final transmitted data

176,160,768 + 7,549,800

= **183,710,568 bytes**

≈ **175.2 MB**

Final Answer

- **Compressed File Size = 168 MB**
- **Segments = 125,830**
- **Protocol Overhead = 7,549,800 bytes**
- **Final Data Transmitted = 183,710,568 bytes (≈175.2 MB)**

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data, provides sequencing, acknowledgements, and error recovery.
- **Data Link Layer:** Creates frames and performs error detection.
- **Network Layer:** Routes packets to the destination.

- **Physical Layer:** Transmits the final bit stream through the communication medium.

Conclusion

The OSI model provides a structured framework for reliable and efficient communication between networked devices. Each layer performs a specific function, from providing user services at the Application Layer to transmitting data through the Physical Layer. Features such as segmentation, flow control, routing, fragmentation, framing, error detection, synchronization, compression, and encryption work together to ensure accurate, secure, and efficient data transmission. Understanding the functions of each OSI layer helps in designing, implementing, and troubleshooting modern computer networks effectively.

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